

Lesson 2C: Solving Equations with Rational Exponents

Criteria	4 - Mastery	3 - Proficiency	2 - Developing	1 - Beginning
Isolation of Term (15%)	Consistently and correctly isolates the term with the rational exponent before applying powers.	Isolates the term with the rational exponent with minor errors.	Attempts to isolate the term but makes significant errors.	Cannot isolate the term with the rational exponent.
Applying Reciprocal Power (25%)	Corrects applies the reciprocal power to both sides to eliminate the exponent.	Applies the reciprocal power correctly but makes minor errors in calculation.	Applies the power to only one side or uses the wrong power.	Cannot identify the need for or value of the reciprocal power.
Handling Even/Odd Roots (20%)	Accurately identifies when a $\pm$ is needed for even roots/exponents and resolves them.	Identifies $\pm$ scenarios but occasionally misses them or uses them incorrectly.	Struggles to differentiate between even and odd rational exponents.	Does not account for potential multiple solutions.
Solving the Resulting Equation (20%)	Accurately solves the resulting polynomial/linear equation.	Solves the resulting equation with minor calculation errors.	Makes significant errors when solving the resulting equation.	Cannot solve the resulting equation.
Extraneous Solutions (15%)	Validates all solutions in the original equation and accurately identifies extraneous solutions.	Checks solutions but makes a mistake in identifying extraneous roots.	Attempts to check solutions but does not understand how to use them.	Does not check for extraneous solutions.
Communication & Precision (5%)	Work is orderly, organized, and written with precise notation.	Work is legible with minor organizational issues.	Work is difficult to follow.	Work is not organized or intelligible.